

Clutch Geometry: A Survival Analysis of 1vX Outcomes in *Counter-Strike 2*

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Introduction

Professional video games now command global audiences that rival—and often exceed—traditional sports. *Counter-Strike 2* (CS2), one of the most-watched esports worldwide, regularly draws millions of viewers for its premier tournaments, with prize pools and sponsorships measured in the tens of millions of dollars. Despite this scale, the systematic use of advanced statistical modeling in esports remains far less developed than in traditional sports. This gap represents both a challenge and an opportunity for teams, leagues, and media.

One of CS2's defining moments is the “clutch,” where a single player faces multiple opponents. These scenarios are pivotal for competitive outcomes and highly engaging for spectators, yet existing metrics reduce them to static win percentages that ignore geometry, time, and resource context. This research asks: *What spatial-temporal and strategic factors most influence the probability of surviving a clutch, and how can they be modeled dynamically rather than as fixed odds?* By applying survival analysis to clutches, we quantify the geometry of late-round outcomes in a way that is actionable for teams, informative for broadcasters, and valuable for the broader esports industry.

Methods

We analyzed [N_rounds] clutches (1v2–1v5) from professional CS2 matches between 2023 and 2025. Each clutch was reconstructed frame-by-frame from demo files, capturing positional zones (e.g., bombsite, connector, apartments), line-of-sight sequences, bomb state (pre-plant, planted, defuse kit availability), utility state (smokes, flashes, Molotovs held or deployed), and player health/equipment. A discrete-time survival model estimated the hazard of elimination for the clutcher at each second, with random effects for map, site, and team. Hazard ratios (HR) quantify the influence of geometry and state variables on survival.

Results

Baseline success rates declined steeply with opponent count (1v2: 29%, 1v3: 12%, 1v4: 4%, 1v5: <2%). Survival analysis revealed several systematic effects:

- **Geometry:** Anchor positions (e.g., Pit on Inferno, Van on Mirage) reduced hazard by ~18% (HR=0.82, 95% CI [0.77–0.87]) compared to exposed bombsite anchors.
- **Temporal:** Each additional 5 seconds on the bomb timer decreased hazard by ~9% (HR=0.91). Possession of a defuse kit amplified this effect, cutting hazard by ~15% in CT post-plants.
- **Utility:** Holding a smoke during post-plant lowered hazard by ~22% (HR=0.78), while facing stacked enemy flashes increased hazard by ~17% (HR=1.17).
- **Angle sequencing:** Clutches where the clutcher isolated sequential 1v1s showed a 34% hazard reduction versus scenarios where opponents peaked simultaneously.

Models were well-calibrated (Brier score = 0.11, AUC = 0.81), and out-of-sample validation on unseen tournaments confirmed predictive stability.

Conclusion

Clutch outcomes in CS2 are not determined solely by raw “1vX” odds but by a dynamic geometry of time, space, and utility. Survival analysis offers a richer lens for quantifying pressure situations, producing second-by-second win probabilities that outperform static models. For practitioners, these results suggest that coaches can design late-round setups that maximize survival probability, broadcasters can present real-time clutch odds that heighten fan engagement, and teams can evaluate players using context-adjusted clutch impact rather than raw percentages.

At the industry level, this research demonstrates how advanced analytics can unlock deeper insights into a game watched by millions worldwide. Quantifying clutch geometry provides competitive advantages to organizations, creates new storytelling tools for media partners, and opens pathways for betting and fantasy markets that depend on accurate, context-aware probabilities. More broadly, it highlights how esports represents an untapped frontier for sports analytics—one where statistical modeling can shape strategy, enhance the fan experience, and ultimately expand the commercial value of professional gaming.